Mid-Submission: Confidence-Based Query Routing for Hybrid Retrieval

1. Project

We test whether cheap per-query features can route each query to BM25, dense, or tuned hybrid retrieval, and escalate low-confidence queries to a cross-encoder under a small budget. Evaluation uses held-out 5-fold predictions and paired bootstrap CIs on NFCorpus, SciFact, FiQA, CQADupStack-Unix and Quora (3,343 queries). The metric is nDCG@10.

2. Hypotheses

Phase 1 (original):

	Hypothesis	Verdict	Evidence
H1	Routing beats hybrid and RRF	Not supported (vs hybrid)	Router −0.0061 [−0.0091, −0.0032]; with τ escalation −0.0054; cost-aware −0.0012 [−0.0025, +0.0002]. Beats RRF by +0.0142, but so does hybrid.
H2	Recall@100 matches hybrid	Supported	All routers within ±0.01.
H3	Budgeted escalation keeps most of the rerank gain	Not supported	At a 10% budget it keeps 16% of the +0.0067 blended gain.
H4	Latency near hybrid	Partly	Router +5–13 ms. 10% escalation is 2.5–7× hybrid. Always-rerank costs 368–406 ms.

Phase 2 (added because Phase 1 failed; exploratory, post-hoc). Natural queries show no routing signal, so we ask when routing can work. We test on a custom skewed set of 288 queries from SciFact and NFCorpus (144 keyword, 144 zero-overlap paraphrase).

- **P2-H1:** routing needs query groups whose best retriever differs; natural sets have none.
- **P2-H2:** the 18 features separate keyword from paraphrase queries, so route labels are learnable.
- **P2-H3:** one shared α is a poor compromise on a mixed-style workload.
- **P2-H4 (planned):** router gain peaks at a balanced style mix and vanishes at either extreme.

3. What we have built

- BM25, dense (e5-small-v2), hybrid and RRF retrievers
- 18 query and retrieval-signal features
- GBM, logistic, advantage-regression and cost-aware routers
- Cross-encoder escalation with predicted-gain, confidence and τ-threshold rules
- Evaluation tooling: oracle headroom, measured latency, risk-coverage analysis, ablations
- Skewed query set and natural-vs-skewed comparison

4. Results so far

Phase 1: always-hybrid (0.5880) is the best dependable cheap policy. The GBM router (0.5819) is not better than random routing at the same proportions (+0.0017 [−0.0010, +0.0044]).

Phase 2 (SciFact + NFCorpus):

	n	Hybrid	Router − hybrid	Oracle gain	AUC (BM25/dense)	Off hybrid
Natural	623	0.536	−0.0007 [−0.005, +0.004]	+0.046	0.64/0.68	5%
Skewed	288	0.589	**+0.0894** [+0.069, +0.111]	+0.120	0.87/0.83	47%

On skewed queries BM25 scores 0.997 on keyword queries and 0.000 on paraphrases, and Recall@100 improves by +0.024 [+0.007, +0.045]. Escalation adds nothing over the router alone (−0.0021 [−0.007, +0.002]), so the gain comes from routing.

Caveat: the skewed set is synthetic and its two styles are separable by construction. It shows routing works under heterogeneous query preference, not that natural workloads contain such queries.

Our next step is to run experiments that validate our new hypotheses.